

Dr. Nathan Grieser

RESEARCH EXPERIENCE

Postdoctoral Research Associate

July 2022 to present

LHCb Experiment; University of Cincinnati, Cincinnati, OH, USA

Advising Principal Investigator: [Conor Henderson](#), Department of Physics

Scientific and Project Leadership

Physics Convener: Quarks, Electroweak, and Exotica Physics Working Group

Deputy Project Leader: Data Processing and Analysis (expected PL Summer 2026)

Convener, LPCC LHC Electroweak Working Group

Steering Board Member, IRIS-HEP

Physics Program Accolades

Designing and executing first measurement of W helicity fractions in the forward region, with precision similar to central measurements in the most BSM-sensitive region

State-of-the-art modeling and theory applications for measurements of fundamental electroweak properties including Z -boson mass and weak mixing angle

Leading the modernization of LHCb Jet-tagging program, applying modern algorithm architectures to increase HF-tagging rates between 20-40%

Infrastructure and Computing Accolades

Scientific oversight of collaboration-wide data processing and analysis workflows

Coordination of legacy data reprocessing and software stack evolution, delivering full Run-2 reprocessing in under 2 months during data-taking

Strategic planning for scalable analysis models in Run 3 and beyond by developing more efficient data formats and resource-use reduction schemes

Chung Yao Chao Fellow

September 2020 to June 2022

LHCb Experiment; Institute of High Energy Physics (IHEP), Beijing, China

Advising Principal Investigator: [Yiming Li](#), Department of Physics

Constructed analysis strategy and software framework for di-parton scattering of B and D mesons measurement

Upstream Tracker Detector control systems and detector installation and commissioning

Design and installed Upstream Tracker Detector Safety System

Led development of B-hadrons and Quarkonia trigger menu for LHCb Run 3

Ph.D. Candidate

August 2014 to August 2020

ATLAS Experiment, High Energy Physics Group

University of Oklahoma, Norman, OK, USA

Advising Professor: [Michael G. Strauss](#), Department of Physics

Analysis Contact: Search for heavy resonances decaying to a pair of W bosons in the fully leptonic decay channel

Software Migration and Selection Optimization for Measurements of the Higgs boson production by gluon-gluon fusion and vector-boson fusion in the $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$ channel

Light jet calibration of flavour-tagging algorithms using the negative tag method

EDUCATION

Ph.D., Physics

August 2014 to August 2020

University of Oklahoma, Norman, OK, USA

Dissertation Advisor: **Michael G. Strauss**Dissertation: *Searches For Heavy Resonances In The $R \rightarrow WW \rightarrow \ell\nu\ell\nu$ Decay Channel Using pp Collisions At $\sqrt{s} = 13$ TeV With The ATLAS Detector At The LHC***B.S., Physics & Astronomy**

August 2010 to May 2013

Ohio Northern University, Ada, OH, USA

SELECTED

PUBLICATIONS

(ANALYSIS LEADER /
CONTRIBUTING
AUTHOR)*Highlighted Publications ([AC = Analysis Contact]). Additionally, I've contributed significantly to 30+ publications as a proponent or convener; a named author in 500+ ATLAS/LHCb papers*

12. [AC] LHCb Collaboration, *In Review Measurement of the Helicity State Fractions of $W^\pm$ Bosons Recorded with the LHCb Run 2 Dataset*. Spring 2026. To be Submitted to Physical Review Letters
11. LHCb Collaboration, *Machine learning techniques for jet reconstruction at LHCb and application to the search for $H \rightarrow b\bar{b}$ and $H \rightarrow c\bar{c}$ in $\sqrt{s} = 13$ TeV pp collisions*. January, 2026. Submitted to Journal for High Energy Physics, <https://arxiv.org/abs/2601.16802>
10. LHCb Collaboration, *Measurement of the $W \rightarrow \mu\nu_\mu$ cross-sections as a function of the muon transverse momentum in pp collisions at 5.02 TeV*. September, 2025. Accepted by Journal of High Energy Physics, [arXiv:2509.18817](https://arxiv.org/abs/2509.18817)
9. [AC] Grieser, *et al.*, *The LHCb Stripping Project: Sustainable Legacy Data Processing for High-Energy Physics*. November, 2025. Computing and Software for Big Science, *Comput Softw Big Sci* 9, 21 (2025)
8. Grieser, *et al.*, *The LHCb Sprucing and Analysis Productions*. June, 2025. Computing and Software for Big Science, *Comput Softw Big Sci* 9, 15 (2025)
7. LHCb Collaboration, *Measurement of the Lund plane for light- and beauty-quark jets*. May, 2025. *Phys. Rev. D* 112 (2025) 072015
6. LHCb Collaboration, *Measurement of the Z-boson Mass*. April, 2025. **Editor's Suggestion** *Phys. Rev. Lett.* 135 (2025) 161802
5. LHCb Collaboration, *Measurement of the Effective Leptonic Weak Mixing Angle*. October, 2024. *JHEP*:12(2024)026
4. ATLAS Collaboration, *Measurements of Higgs boson production by gluon-gluon fusion and vector-boson fusion using $H \rightarrow WW^* \rightarrow e\mu e\nu$ decays in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*. August, 2023. *Physical Review D*:108.032005
3. [AC] ATLAS Collaboration, *Search for heavy resonances in the decay channel $W^+W^- \rightarrow e\mu e\nu$ in pp Collisions at $\sqrt{s} = 13$ TeV using 139 fb^{-1} of data with the ATLAS detector*. November, 2022. ATLAS Note:ATLAS-CONF-2022-066
2. Grieser, *et al.*, *The SALT – Readout ASIC for Silicon Strip Sensors of Upstream Tracker in the Upgraded LHCb Experiment*. December, 2021. *MDPI Sensors*:10.3390/s22010107
1. ATLAS Collaboration, *Calibration of light-flavour b-jet mistagging rates using ATLAS proton-proton collision data at $\sqrt{s} = 13$ TeV*. April, 2018. ATLAS Note:ATLAS-CONF-2018-006

SELECTED INVITED
TALKS AND
HIGH-PROFILE
PRESENTATIONS

11. **Invited Seminar** “Probing the Electroweak Sector and Partonic Structure in the Forward Region with LHCb”, Massachusetts Institute of Technology LNS seminar, Boston, Massachusetts, USA, October 28, 2025.
10. “**LHCb Highlights**”, HIGGS 2025 Plenary, Brown University, Providence, Rhode Island, USA, October 27, 2025.

9. **Invited Seminar** “Recent LHCb results in precision electroweak physics, and outlook for the upcoming Run 3 data set”, University at Buffalo Physics Seminar, Buffalo, New York, USA, August 26, 2025.
8. “QCD, electroweak physics, and searches for exotic signatures in the forward region at LHCb”, Moriond QCD Plenary, La Thuile, Italy, March 30, 2025.
7. “Heavy-Flavour Jet Tagging at LHCb Using Graph Neural Networks”, ML4Jets, LPNHE, Paris, France, November 5, 2024.
6. “W Mass, Weak Mixing Angle, and W Cross Sections”, Implications of LHCb Measurements and Future Prospects Plenary, CERN, Geneva, Switzerland, October 25, 2024.
5. “LHCb Stripping Project: Continuing to Fully and Efficiently Utilize Legacy Data”, Conference on Computing in High Energy Physics, Krakow, Poland, October 21, 2024.
4. **Invited Seminar** “Beyond Flavour: Exploring unique measurements of electro-weak, Higgs, and exotica with the LHCb detector”, University of Maryland HEP / Particle Astro seminar, College Park, Maryland, USA, April 12, 2024.
3. “Physics with W and Z Bosons at the LHCb Experiment”, European Physical Society Conference on High Energy Physics, Hamburg, Germany, August 23, 2023.
2. “Python Usage Within the LHCb Experiment”, PyHEP Workshop 2022, Online, September 14, 2022.
1. “Analysis user experience with Python HEP data science tools in LHCb”, IRIS-HEP Analysis Grand Challenge Workshop, Online, April 26, 2022.

SCIENTIFIC
LEADERSHIP
*Summary of
significant leadership
roles within
LHCb/ATLAS*

Deputy Project Leader (*Ex-Legacy software package coordinator*) Winter 2024 to present
LHCb Data Processing and Analysis Project

Convener (*Ex-Sub-Convener SM subgroup*) Fall 2023 to present
LHCb Quarks, Electro-weak, and Exotica Physics Working Group

Detector Safety System Package Leader Spring 2021 to Fall 2024
LHCb Upstream Tracker Detector

$WW \rightarrow \ell\nu\ell\nu$ Contact Spring 2019 to Fall 2021
ATLAS Heavy Resonance Combination

Physics Modelling Group Contact Fall 2019 to Fall 2020
ATLAS Higgs to WW Analysis Group

LHC-WIDE
COMMUNITY
LEADERSHIP AND
SERVICE

Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP) January 2025 to present

Steering Board Member

LPCC LHC Electroweak Working Group Winter 2024 to present
Co-Convener

HIGGS 2026 October 2026
Program Committee

HIGGS 2025 October 2025
BSM Higgs session co-convener

Conference on Computing in High Energy and Nuclear Physics (CHEP) October 2024
Collaborative software and maintainability session co-convener

Large Hadron Collider Physics Conference (LHCP) June 2024
Electro-weak session co-convener

Washington, D.C. High Energy Physics Advocacy Trip April 2024
US LHC Users Association Young Physicists Representative

HONORS, AWARDS, AND FUNDING	IRIS-HEP Post-doctoral Support for research on Analysis Systems 0.5 FTE Salary for one year	2025
--------------------------------	---	------

US LUA Lightning Talk Prize Winner First Measurement of W Helicity Fractions at LHCb	2023
---	------

Chung-Yao Chao Fellowship	2021
---	------

STUDENTS MENTORED	Kylee Monaghan	June 2025 to August 2025
----------------------	---------------------------------------	--------------------------

<i>Undergraduate students supervised in projects which I designed</i>	Projects advised on: Di-Jet Angular Analysis as a Probe for New Physics Awards: 2026 Rhodes Scholar LHCb Summer Student Moritz Karbach Prize Position during mentorship: CERN Summer Student Current position: Undergraduate student at Sewanee: The University Of The South, Tennessee, USA
---	---

Gabriella Pesticci	June 2024 to July 2025
---	------------------------

Projects advised on: Jet Flavour Classification with a Graph Neural Network (GNN) at the LHCb Awards: US LHC Users Association Lightning Round Winner Position during mentorship: CERN Summer Student, Undergraduate capstone researcher Current position: PhD Candidate in Particle Physics at Michigan State University, Michigan, USA
--

Carrie Cox	June 2023 to August 2023
-----------------------------------	--------------------------

Projects advised on: Phenomenological study of t-channel production of leptoquarks Position during mentorship: CERN Summer Student Current position: Studying M.A. in accelerator science and engineering at Stony Brook University
